A GUI for Magboltz

IFIN-HH (Romania)

7th DRD1 Collaboration Meeting
WG4 — Modeling and Simulations

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Magboltz in Detector Physics

- Monte Carlo electron transport simulation
- Fundamental tool for:
  - Gas detector design
  - Drift properties
  - Diffusion
  - Townsend / attachment
- Used across:
  - MPGDs
  - TPCs
  - RPC studies
  - Simulation chains (Garfield++)

Magboltz remains a cornerstone of gas detector modeling.

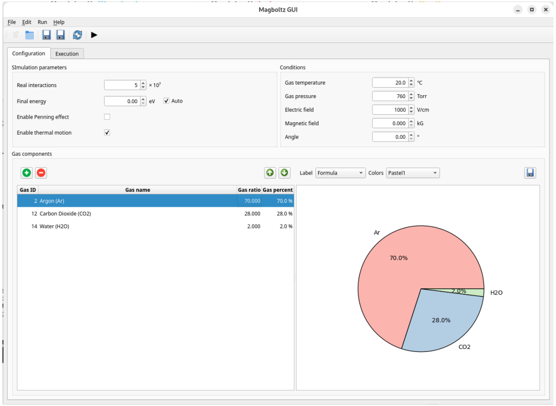
The CLI Workflow Reality

- Text-based input cards
- Text-based output parsing
- Typical friction points:
  - Input syntax errors
  - Iterative parameter tuning
  - Manual data extraction
  - Ad-hoc plotting scripts
  - Student onboarding difficulty

Powerful physics engine, but workflow-heavy interface.

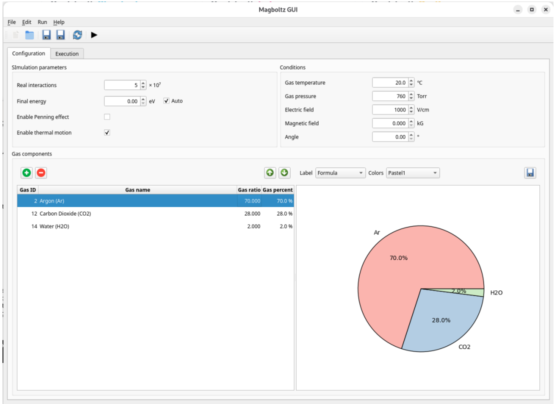
Magboltz GUI

- Python + PyQt6 application
- Cross-platform
- Designed as a workflow layer, not a replacement
- Core functions:
  - Input file creation & editing
  - Magboltz execution wrapper
  - Output parsing → structured model
  - Data export → CSV / JSON / XML
  - Built-in plotting tools



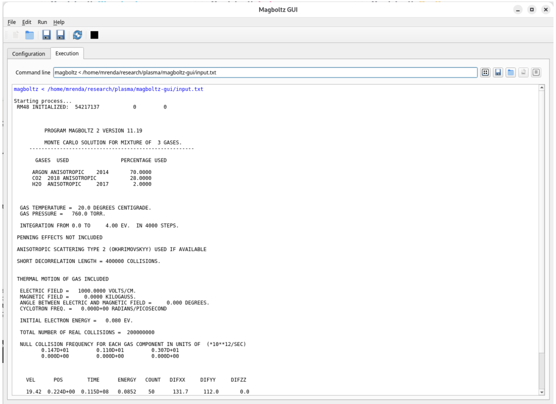
Simulation & Gas Configuration

- Editable simulation parameters
- Gas mixture editor
- Ratio $\rightarrow$ automatic normalization
- Visual mixture representation
- Reduced input mistakes during iteration



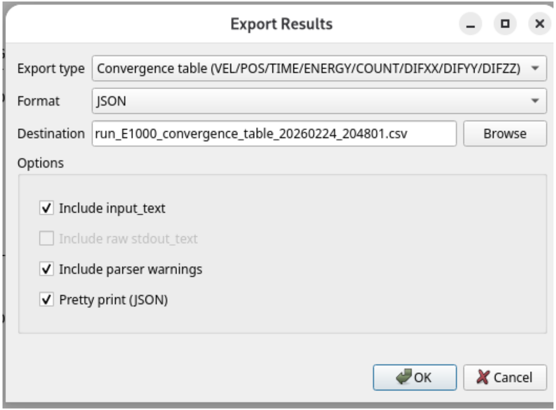
Execution Workflow

- Direct Magboltz invocation
- Command line transparency
- Live stdout/stderr streaming
- Run/Stop controls
- Raw output persistence
- Reopen previous runs

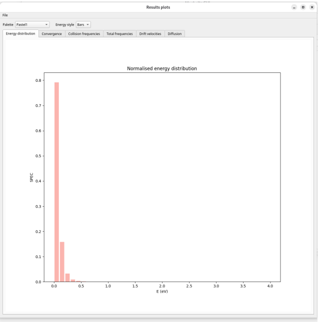


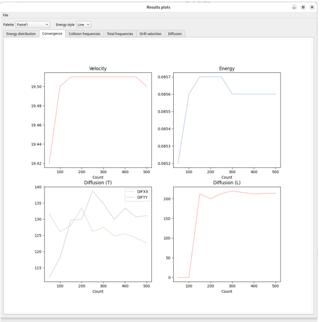
From Text Output → Structured Data

- Canonical RunResult model
- Extracted:
  - Transport coefficients
  - Collision frequencies
  - Convergence tables
  - Energy distributions
- Export formats:
  - CSV
  - JSON
  - XML
- Reproducible analysis pipelines



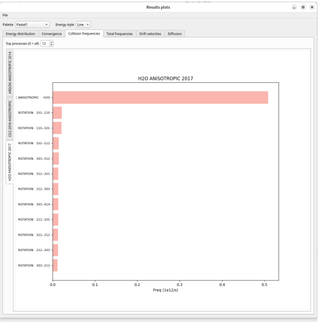
Built-in Plotting Tools

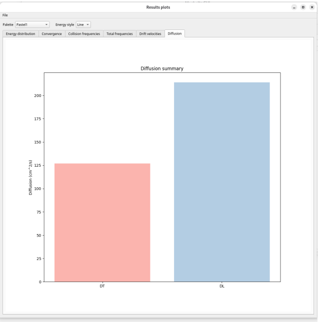




Immediate visual feedback without external scripts.

Built-in Plotting Tools (cont.)





Immediate visual feedback without external scripts.

Availability & License

Project repository

- GitLab: `https://gitlab.com/micrenda/magboltz-gui/`
- License: MIT
- Documentation: README in the repository

Setup: pip (bundled Qt)

Recommended when you want Qt bundled with the app.

- Python environment
- Magboltz binary available in PATH

Commands:

```
pip install "magboltz-gui[qt]"
magboltz-gui
```

Setup: pip (system Qt)

Recommended on Linux when using system PyQt6.

- Install system Qt bindings
- Magboltz binary available in PATH

Commands (Debian/Ubuntu):

```
sudo apt install python3-pyqt6
pip install magboltz-gui
magboltz-gui
```

In Memoriam

Stephen F. Biagi

(1949–2025)

We dedicate this work to his memory.

His pioneering contributions to modeling electron transport in gases profoundly shaped detector physics.

This interface is a modest tribute to his legacy.